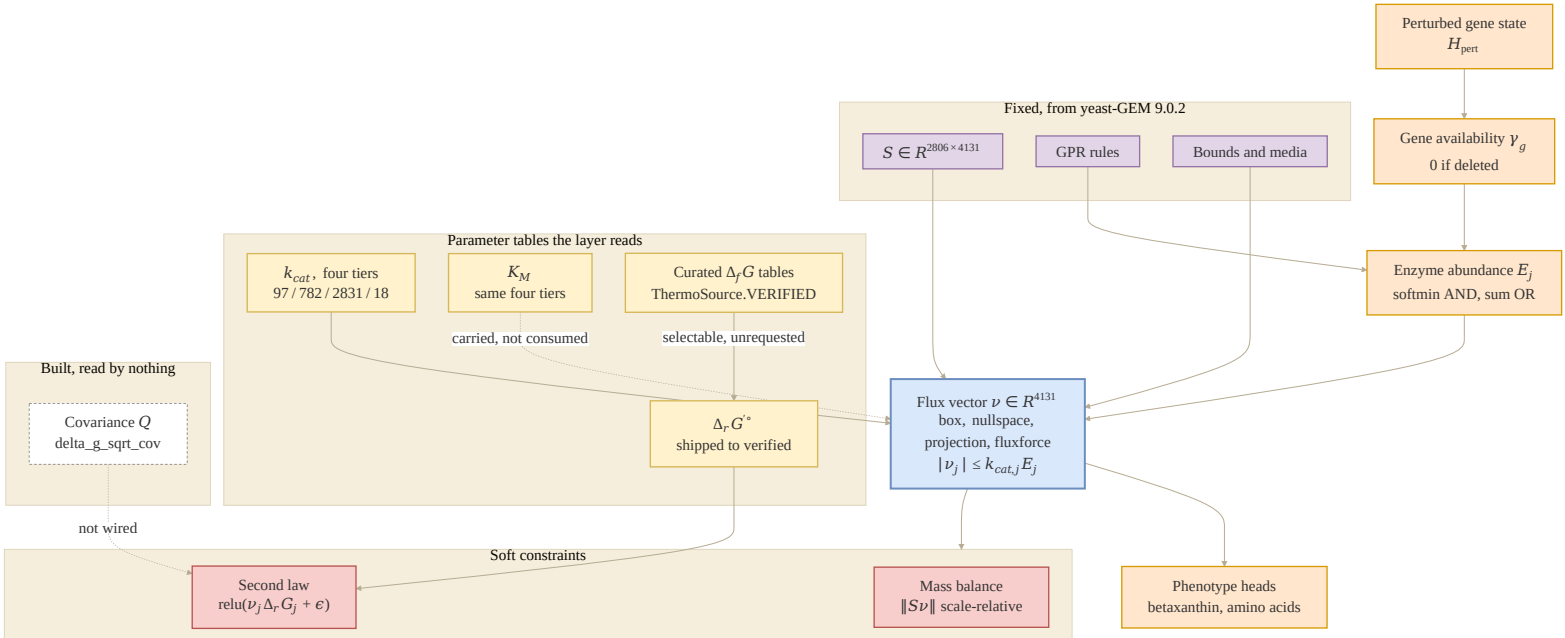

Perturbed gene state
H_{pert}
Fixed, from yeast-GEM 9.0.2
$S \in R^{2806 \times 4131}$
GPR rules
Bounds and media
Gene availability γ_g
0 if deleted
Parameter tables the layer reads
k_{cat}, four tiers
97 / 782 / 2831 / 18
K_M
same four tiers
Curated $\Delta_f G$ tables
ThermoSource.VERIFIED
Enzyme abundance E_j
softmin AND, sum OR
carried, not consumed
selectable, unrequested
Built, read by nothing
Covariance Q
delta_g_sqrt_cov
$\Delta_r G'^{\circ}$
shipped to verified
Flux vector $v \in R^{4131}$
box, nullspace,
projection, fluxforce
$|v_j| \le k_{cat,j} E_j$
not wired
Soft constraints
Second law
relu($v_j \Delta_r G_j + \epsilon$)
Mass balance
$\|Sv\|$ scale-relative
Phenotype heads
betaxanthin, amino acids